

# Terrence Alsup

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## Experience

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### Morgan Stanley

*Vice President, Machine Learning Researcher*

*New York, NY*

05/2025 – Present

- Led development of a U.S. residential AVM, supporting LTV calculations for Securitized Products, Lending, and Credit Risk teams
- Developed a load forecasting model for day-ahead power markets using tabular foundation models for commodity traders
- Partnered with muni e-trading strats to develop a real-time estimation framework for the MMD index curve

### Finastra

*Senior Data Scientist*

*Remote*

12/2023 – 05/2025

- Partnered with the Chief AI Officer to establish Finastra's AI strategy
- Developed ReAct agent with LangGraph for natural language driven data analytics platform

*Data Scientist*

02/2022 – 11/2023

- Developed a multi-period churn prediction model for bank customers using XGBoost and isotonic regression
- Improved fidelity of StyleTTS2, an open-source diffusion-based text-to-speech model, for narration in product training videos

### Sandia National Laboratories

*Computer Science Research Institute Intern*

*Remote*

06/2021 – 09/2021

- Worked with Tommie Catanach on applications of Bayesian optimal experimental design to sensor configuration for seismic monitoring
- Developed a method for estimating the error in expected information gain estimates

## Education

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### Courant Institute of Mathematical Sciences, New York University

Ph.D. Mathematics

Advisor: Benjamin Peherstorfer

Thesis: *Trading Off Deterministic Approximations and Sampling in Multi-fidelity Bayesian Inference*

*New York, NY*

09/2017 – 01/2023

### Georgia Institute of Technology

M.S. Mathematics

B.S. Applied Mathematics

*Atlanta, GA*

01/2017 – 08/2017

09/2013 – 12/2016

## Awards

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Vista Hackathon Winner - The Grid: Federated Learning for the Credit Invisible

2022

BGCE Student Paper Prize Finalist

2021

SIAM Student Travel Award

2020, 2022

Henry MacCracken Fellowship

2017 – 2023

STEM Fellowship (Georgia Tech Math Department)

2017

## Publications

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- [1] A. Maurais, **T. Alsup**, B. Peherstorfer, and Y. Marzouk. Multi-fidelity Covariance Estimation via Regression on the Manifold of Symmetric Positive Definite Matrices. *SIAM Journal on Mathematics of Data Science*, 2025.
- [2] **T. Alsup**, T. Hartland, B. Peherstorfer, and N. Petra. Further Analysis of Multilevel Stein Variational Gradient Descent with an Application to the Bayesian Inference of Glacier Ice Models. *Advances in Computational Mathematics*, 2024.
- [3] A. Maurais, **T. Alsup**, B. Peherstorfer, and Y. Marzouk. Multi-fidelity Covariance Estimation in the Log-Euclidean Geometry. In *International Conference on Machine Learning (ICML)*, 2023.
- [4] **T. Alsup** and B. Peherstorfer. Context-Aware Surrogate Modeling for Balancing Approximation and Sampling Costs in Multi-fidelity Importance Sampling and Bayesian Inverse Problems. *SIAM/ASA Journal on Uncertainty Quantification*, 2022.
- [5] **T. Alsup**, L. Venturi, and B. Peherstorfer. Multilevel Stein Variational Gradient Descent with Applications to Bayesian Inverse Problems. In *Mathematical and Scientific Machine Learning (MSML) 2021*, 2021.
- [6] **T. Alsup** and T. Catanach. Expected Information Gain Estimates and Bayesian Optimal Experimental Design. In J.D. Smith and E. Galvan, editors, *Computer Science Research Institute Summer Proceedings 2021*, pages 269–282, 2021. Technical Report: SAND2022-0653R.

## Teaching

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|--------------------------------------------------------------------------------|-----------|
| Mathematical Statistics, <i>NYU</i> , Teaching Assistant (3 semesters)         | 2020–2021 |
| Introduction to Machine Learning, <i>NYU</i> , Class Assistant (2 semesters)   | 2019–2020 |
| Stochastic Calculus, <i>NYU</i> , Teaching Assistant (1 semester)              | 2018      |
| Differential Equations, <i>Georgia Tech</i> , Teaching Assistant (2 semesters) | 2014–2015 |

## Projects

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|-----------------------------------------------------------------------------------|------|
| Fantasy Football Luck Calculator for Round-Robin Scheduling                       | 2024 |
| Learning Large Graphical Models via Convex Optimization                           | 2020 |
| Parallel Kinetic Monte Carlo, <i>with Anya Katsevich</i>                          | 2019 |
| Multi-fidelity Cross-Entropy Estimation of Rare Events, <i>with Frederick Law</i> | 2019 |
| Image Forgery Detection in Alipay, <i>with Ant Financial as a part of RIPS-HK</i> | 2016 |

## Technical Skills

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|------------------------------|---------------------------------------------------------|
| <b>Programming Languages</b> | Python, Matlab, Q, SQL                                  |
| <b>Databases</b>             | KDB+, Relational databases, Vector databases            |
| <b>Deep Learning/AI</b>      | Pytorch, Lightning, LangGraph, OpenAI APIs, Claude Code |
| <b>HPC</b>                   | MPI, Multiprocessing, Slurm, PySpark                    |
| <b>Cloud</b>                 | Databricks, Azure                                       |